

Solar System Sizing & Recommendation Engine Logic

Technical Reference Document

1. Required User Inputs

The following variables must be collected from the user via the frontend interface to perform the calculations:

- **Monthly Electricity Consumption (*Monthly Units*):** Measured in kWh. (Required)
- **Desired Backup Hours (*Backup Hours*):** Number of hours the system must sustain the load during a grid outage. (Optional, default = 0)

2. Engineering Constants

The following values are established engineering constants for the deployment context and are hardcoded into the sizing engine:

- **Peak Sun Hours (H_{PSH}):** 4.5 hours/day (Average practical irradiance).
- **Performance Ratio (PR):** 0.80 (Accounts for temperature, inverter, and wiring losses).
- **Standard Panel Wattage (*Panel Wattage*):** 550 Watts (Standard Tier-1 module).
- **Inverter Efficiency (η_{system}):** 0.95 (Standard hybrid inverter efficiency).
- **Battery Depth of Discharge (DoD):** 0.80 (Standard LFP Lithium operational limit).
- **Minimum System Size:** 3.0 kW (Minimum viable residential installation).
- **Panel Area Footprint:** 30 sq. ft. per 550W module (Includes mounting clearances).

3. Mathematical Formulas

Formula 1: Daily Energy Consumption

Converts the monthly billing units into a daily average.

$$E_{daily} = \frac{Monthly\ Units}{30} \quad (1)$$

Formula 2: Preliminary PV Array Sizing

Calculates the required DC solar capacity in kilowatts (kW).

$$P_{PV} = \frac{E_{daily}}{H_{PSH} \times PR} \quad (2)$$

Formula 3: Panel Count Calculation

Determines the physical number of solar panels required.

$$Panel\ Count = \lceil \frac{P_{PV} \times 1000}{Panel\ Wattage} \rceil \quad (3)$$

Formula 4: Roof Area Estimation

Calculates the minimum required unobstructed rooftop space.

$$Required\ Area = Panel\ Count \times 30 \quad (4)$$

Formula 5: Battery Capacity Sizing

Executed only if *Backup Hours* > 0. Calculates the required storage capacity in kWh.

$$C_{battery} = \frac{(E_{daily}/24) \times Backup\ Hours}{\eta_{system} \times DoD} \quad (5)$$

4. System Output Data Structure

The recommendation function must return a structured dictionary containing the computed variables, which will be passed to the LLM and the frontend.

- `recommended_system_kw`: The calculated P_{PV} (float, rounded to 2 decimal places). Minimum floor of 3.0.
- `recommended_panels_count`: The calculated *Panel Count* (integer).
- `panel_wattage`: 550 (integer).
- `required_roof_area_sqft`: The calculated *Required Area* (integer).
- `battery_capacity_kwh`: The calculated $C_{battery}$ (float, rounded to 2 decimal places, or 0 if no backup requested).
- `disclaimer`: "This is an initial estimate and does not replace a professional site survey or engineering design." (string).